

The Breast Cancer Gene Network as a Complex Adaptive System: Controllability, Feedback Structure and Relapse-Free Survival in METABRIC, with Cross-Cohort Transportability Assessment in TCGA-BRCA

[Anonymised for double-anonymized review; author and affiliation are provided on the title page]

Running title: Cybernetic control-kernel score and breast cancer prognosis

Keywords: breast cancer; gene association network; controllability; control energy; survival analysis; graphical lasso

Abstract

Background: Cancer progression is often described as the behaviour of a complex adaptive system; we ask whether control-theoretic descriptors of its transcriptional association network carry incremental prognostic information beyond conventional clinical variables. **Methods:** From 1,979 METABRIC primary tumours we inferred a sparse 99-gene conditional-dependence network, computed cybernetic descriptors and built a patient-level control-kernel score (CS) with structure-based, outcome-blind weights. CS was tested against relapse-free survival (RFS) by Cox, repeated and full-pipeline cross-validation, log-rank, permutation, Bayesian horseshoe and Schoenfeld proportional-hazards analyses; cross-cohort transportability was assessed in TCGA-BRCA ($n = 710$) on overall survival (OS). **Results:** The inferred network is cycle-rich (242 independent undirected cycles), stable under dissipative realisation (spectral radius 0.904, damping $\eta = 1.2$), and requires 24 independent driver channels (75-dimensional controllable subspace). CS was associated with RFS after adjustment (HR 1.22 per SD, $p = 5.6 \times 10^{-6}$), with strongly time-dependent effect (Schoenfeld $p = 2.0 \times 10^{-14}$): HR per SD 1.65 within 0–24 months, 1.29 within

24–60, 0.91 after 60. Full-pipeline cross-validation improved the C-index from 0.587 ± 0.041 (clinical) to 0.606 ± 0.031 (+CS); on the fixed network, 0.588 to 0.611. In TCGA-BRCA, CS showed no association with OS (HR 1.00, 95% CI 0.80–1.25, $p = 0.99$), and the inferred network did not reproduce across cohorts (Jaccard index 0.02–0.04). **Conclusions:** A score derived from the network’s structural control kernel carries prognostic information for RFS in METABRIC, but the association is cohort- and endpoint-specific and time-dependent; an observational conditional-dependence network should not be interpreted as a causal, portable regulatory programme.

1 Introduction

Since Wiener’s formulation of cybernetics as the science of control and communication in animals and machines (Wiener 1948), the language of feedback, stability and control has proved remarkably productive in biology (Thomas et al. 2019). Gene regulatory networks (GRNs) are prototypical cybernetic objects: their function emerges from dense feedback wiring, their operating point is maintained by homeostatic (negative-feedback) circuits, and their collective behaviour can be analysed with the tools of linear and nonlinear control theory (Liu and Barabási 2016; Zañudo et al. 2017). At the same time, cancer is increasingly framed as a complex adaptive system—a multilevel, self-organising network whose phenotypic states correspond to stable attractors of the underlying molecular dynamics (Li and Balázsi 2018; Feinberg and Levchenko 2023; Shin and Cho 2023; Berner et al. 2023). In this view a tumour is not a fixed molecular entity but a dynamical regime of a regulatory network, and recurrence or resistance can be interpreted as transitions between attractors.

Breast cancer is a natural test bed for this systems-level perspective. Large transcriptomic cohorts such as METABRIC (Curtis et al. 2012) have made it possible to reconstruct condition-specific gene networks from patient-level expression data and to link network structure to clinical outcome (Nersisyan et al. 2021; Wang and Liu 2022; Huo et al. 2022). A recurring finding is that the topology of the inferred network—rather than any single gene—carries prognostic sig-

nal: rewiring analyses (Wang and Liu 2022) and network-driven survival analyses of breast cancer transcriptomes (Nersisyan et al. 2021; Huo et al. 2022) associate network-level features with outcome, while Boolean-model studies characterise the attractor structure of the inferred regulatory programme (Sgariglia et al. 2021). Established multigene expression signatures already translate gene-expression measurements into prognosis (Hatzis et al. 2011), but what is missing is a direct link from the control-theoretic descriptors of network structure—controllability, control energy and structural cycle richness (Liu et al. 2011; Yuan et al. 2013; Gao et al. 2014; Yan et al. 2015)—to patient-level survival models. This study provides such a link by translating a purely structural control analysis of the breast cancer GRN into a patient-level prognostic score. Throughout, we use the term *GRN* as a computational shorthand for the inferred gene-level conditional-dependence network, not as a claim of a causal, experimentally verified regulatory map.

An important epistemic distinction frames the analysis and its interpretation. (i) The network is an *observational conditional- dependence structure* estimated from cross-sectional expression; it does not claim to be a causal, experimentally verified regulatory programme. (ii) The network descriptors are *structural properties* of that estimated object—cycle-space dimension, spectral stability, controllability and control-energy budgets—computed without any access to clinical outcome. (iii) The patient score is a *control-energy-inspired summary* of expression projected onto the network’s control kernel. (iv) Any association of the score with survival is an *empirical prognostic association* whose generalisability must be demonstrated, not assumed. We keep these four levels separate throughout: in particular, we do not claim that the inferred network is the physical control system of the cell, nor that the score is a causal determinant of outcome. Cross-cohort assessment in an independent cohort is treated as a decisive test of the fourth level.

In this paper we address three questions. (i) What is the cybernetic organisation of the breast cancer GRN inferred from a large expression cohort? (ii) Viewed as a complex adaptive system, is the inferred network dynamically stable, and does its dynamical regime support a multistable-attractor view of tumour phenotypes or a single, homeostatic operating point? (iii) Does a patient-level score derived from the control-kernel structure, computed without access to clinical outcome,

carry prognostic information for relapse-free survival (RFS) after adjustment for standard clinical covariates—and does that association transport to an independent cohort?

2 Materials and Methods

2.1 Data and Pre-processing (METABRIC)

We analysed mRNA expression and clinical data from the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) cohort (Curtis et al. 2012; Rueda et al. 2019), a collection of primary invasive breast tumours with long-term follow-up. Expression values were downloaded from the cBioPortal data repository (study `brca_metabric`). Following quality control, 1,979 tumours with complete expression data for 99 a-priori breast-cancer regulatory genes were retained. The gene set spans the principal regulatory axes of breast cancer biology: oestrogen receptor signalling (ESR1, PGR), epidermal growth factor receptor signalling (EGFR, ERBB2/HER2), PI3K/AKT/mTOR signalling, cell-cycle control (CDK1, CCNA2, CCNB1, CCNB2, CCND1/2, CCNE1, MYC), the p53/DNA-damage response (TP53, MDM2), apoptosis and EMT regulators, and core transcription factors (Supplementary Table S1). Per-gene expression was z-score standardised across samples. Clinical covariates comprised oestrogen receptor (ER), progesterone receptor (PR) and HER2 status, histological grade and age at diagnosis (PR status missing for two patients and HER2 status for one; Table 1). The primary endpoint was relapse-free survival (RFS), defined as the time to any invasive relapse event or last follow-up.

2.2 Conditional-Dependence Network Inference

We inferred a sparse, undirected conditional dependence network among the 99 genes with the graphical lasso (Friedman et al. 2008), which estimates the precision (inverse covariance) matrix Θ by

$$\hat{\Theta} = \arg \max_{\Theta \succ 0} \{ \log \det \Theta - \text{tr}(\mathbf{S}\Theta) - \alpha \|\Theta\|_1 \}, \quad (1)$$

where $\mathbf{S}$ is the sample covariance of the z-scored expression matrix and the penalty α was selected by 3-fold cross-validation over six candidate values. Partial correlations were derived as $\pi_{ij} = -\hat{\Theta}_{ij} / \sqrt{\hat{\Theta}_{ii}\hat{\Theta}_{jj}}$; an edge (i, j) was declared present when $|\pi_{ij}| > 0.05$ for the sparse network, and the full weighted graph used the absolute partial correlations above 10^{-12} for the topological and control analyses. Edge stability was assessed by re-estimating the network on 25 bootstrap resamples (80% of samples each) and recording the fraction of resamples in which each inferred edge reappeared.

2.3 Cybernetic Network Metrics

Let $G = (V, E)$ be the inferred GRN with $N = |V|$ genes, $E = |E|$ edges and C connected components. Three cybernetic descriptors were computed. (i) *Cycle-space dimension*: the cyclomatic number $M = E - N + C$ counts the independent undirected cycles of the graph. In an undirected conditional-dependence network these cycles do not by themselves identify directed regulatory feedback loops; they measure the structural capacity of the graph to support closed-loop control were directionality known. (ii) *Spectral stability*: for the weighted adjacency matrix $\mathbf{A}$ with entries $|\pi_{ij}|$, the local stability of the linearised dissipative dynamics $\dot{\mathbf{x}} = (\mathbf{A} - \eta\mathbf{I})\mathbf{x}$ is governed by the spectrum of $\mathbf{A}$; choosing dissipation $\eta > \rho(\mathbf{A})$ places the spectrum in the left half-plane. (iii) *Controllability*: for the linear dynamics $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$, the minimum number of independent input channels equals the largest geometric multiplicity of any eigenvalue of $\mathbf{A}$ (Yuan et al. 2013); for the symmetric matrix inferred here this maximum multiplicity is the $N - \text{rank}(\mathbf{A}) = 24$ -dimensional zero eigenspace. As a structure-based proxy for the nodes most relevant to steering the network state, we ranked genes by a combined control score c_i equal to the mean of the percentile ranks of degree, betweenness centrality and eigenvector centrality; the *control kernel* was defined as the top 60 ranked genes. This kernel is a centrality-based structural proxy and is distinct from the minimum driver set of the exact-controllability analysis; the two notions are kept separate throughout.

2.4 Control Energy

We modelled the linearised, dissipative network dynamics

$$\dot{\mathbf{x}}(t) = (\mathbf{A} - \eta \mathbf{I}) \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t), \quad (2)$$

with dissipation $\eta = 1.2 > \rho(\mathbf{A})$ and input matrix $\mathbf{B}$ selecting the control-kernel genes. The controllability Gramian $\mathbf{W}_c$ solves the continuous Lyapunov equation $\mathbf{A}_\eta \mathbf{W}_c + \mathbf{W}_c \mathbf{A}_\eta^\top = -\mathbf{B} \mathbf{B}^\top$, with $\mathbf{A}_\eta = \mathbf{A} - \eta \mathbf{I}$. Because the kernel inputs drive only 75 of the 99 state dimensions, $\mathbf{W}_c$ is singular; we report its spectrum in the reachable subspace, truncate eigenvalues below 10^{-12} for display, and state the dimension of the controllable subspace explicitly. The minimum energy required to steer the system from the origin to a target state $\mathbf{x}_f$ in the controllable subspace is $E_{\min} = \mathbf{x}_f^\top \mathbf{W}_c^+ \mathbf{x}_f$, where the pseudo-inverse $\mathbf{W}_c^+$ is evaluated numerically as the inverse of the regularised Gramian $\mathbf{W}_c + \delta \mathbf{I}$, with $\delta = 10^{-6} \text{tr}(\mathbf{W}_c)/N$ (here $\delta = 2.7 \times 10^{-7}$); the pseudo-inverse is recovered in the limit $\delta \rightarrow 0$. We report the eigenvalue spectrum of $\mathbf{W}_c$, its condition number, its effective rank (number of eigenvalues above 10^{-6} of the maximum) and the fraction of the total Gramian trace carried by the top 10% of modes, which quantifies how concentrated the inexpensive control channels are (Yan et al. 2015). For each kernel gene we also computed the trace of the single-input Gramian as a per-driver measure of controllability.

2.5 Dynamical Simulation and Convergence Analysis

The inferred network was treated as a complex adaptive system in two complementary simulation regimes. (i) *Linear dissipative regime*: we integrated Eq. (2) from 40 random initial states and projected the terminal states onto the Perron eigenvector of $\mathbf{A}$ (normalised to unit sum), obtaining a scalar phenotype coordinate; the projections at the horizon were split at the sample median into low- and high-response halves that define two response poles used for the constant-input perturbation experiment. (ii) *Nonlinear saturating regime*: we integrated $\dot{\mathbf{x}} = -\mathbf{x} + \tanh(\mathbf{A}\mathbf{x}) + \mathbf{B}\mathbf{u}$ from random initial states (drawn independently for the two regimes) and partitioned the terminal

projections into three bands by tertiles to characterise transient spread. To test whether these bands correspond to distinct attractors, we located fixed points of the nonlinear system by fixed-point iteration and multi-start root finding and assessed their local stability from the Jacobian spectrum. In addition, we ran a constant-input perturbation experiment: starting from the state with the lowest transient projection, we applied the constant input $\mathbf{u}_{\text{const}} = \mathbf{B}^\top \mathbf{W}_c^+ \Delta$ (regularised Gramian; Δ the difference between the lowest and highest projection states), integrated the linearised dynamics over a horizon of $T = 8$, and compared the resulting trajectory with the uncontrolled trajectory from the same initial state. We report the integrated input energy $\int_0^T \|\mathbf{u}\|^2 dt$, the endpoint projection, the full-state error relative to the target, and whether the uncontrolled run also crosses the median projection line. This experiment is presented as a demonstration of input-modulated transient response only; it is not claimed to be a minimum-energy steering or a phenotype-transfer result.

2.6 Control-Kernel Patient Score

For each patient s we defined the control-kernel score

$$\text{CS}_s = \frac{\sum_{k \in \mathcal{K}} c_k z_{sk}}{\sum_{k \in \mathcal{K}} c_k}, \quad (3)$$

where $\mathcal{K}$ is the control kernel, z_{sk} is the z-scored expression of gene k in patient s , and c_k is the structure-based, outcome-blind control score of gene k . Crucially, c_k is computed from the network topology only, so CS_s contains no information about survival or recurrence. The control-energy analysis of Section 2.4 motivates the kernel as a set of structurally influential genes; the score itself uses only the centrality weights and no dynamic or energy quantities, so that the prognostic analysis tests the structural hypothesis rather than a fitted dynamical model. The kernel size $K = 60$ was pre-specified as the top K genes by the control score before any model fitting; the sensitivity of the association to K is reported in Section 3.6. The score was standardised across patients (mean 0, unit SD) for interpretation.

2.7 Survival Models and Inference (METABRIC)

All models were estimated on the 1,979 patients with complete RFS and expression data. The Cox proportional-hazards model (Cox 1972) was fitted by maximising the partial likelihood (Breslow tie handling) with an analytic gradient and BFGS optimisation; standard errors were obtained from the BFGS quasi-Newton approximation to the inverse Hessian of the negative partial log-likelihood, and the Harrell concordance index (C-index) was used for discrimination. Grade and age were z-scored across the analysis sample, and the 88 missing grade values were imputed to the z-score mean (zero). As an independent check, the Cox coefficients, standard errors and confidence intervals were re-estimated with the R *survival* package (R 4.5.2, survival 3.8.3, Efron tie handling), which computes the observed information matrix exactly, and agreed with the Python estimates at the reported precision. Three nested specifications were compared: M0 (clinical covariates only: ER, PR, HER2, grade, age), M1 (M0 + CS) and M2 (Ridge-Cox on the top 40 kernel-gene expressions plus clinical covariates, ridge penalty 0.08). Discrimination was evaluated with repeated 5-fold cross-validation (5 repeats, 25 splits in total, stratified by the event indicator), reporting mean and SD of the C-index. Because the network, the kernel genes, the CS weights and the clinical encoding are constructed once on the full cohort in the main analysis, we additionally ran a full-pipeline 5-fold cross-validation in which every feature-construction step (expression standardisation, graphical-lasso inference, centrality computation, kernel selection, CS computation and clinical encoding) was refitted inside each training fold and applied to the held-out fold; this second protocol measures generalisation of the complete pipeline to unseen patients. Survival differences between CS-high and CS-low strata (median split) were tested with the log-rank test (Mantel–Cox form, with the standard variance estimator) and displayed as Kaplan–Meier curves. The survival association of CS was additionally tested with a permutation test in which the CS vector was permuted 1,000 times while the survival times and event indicators were held fixed, the univariate Cox model for CS was refitted on each permuted score, and the observed CS coefficient was compared with its permutation distribution; the reported p-value uses the finite-resolution correction $(r + 1)/(B + 1)$. Finally, as a secondary exploratory analysis, we fitted a

Bayesian logistic model of 5-year relapse with a horseshoe prior on the kernel-gene coefficients (Polya–Gamma Gibbs sampler, 4,000 iterations, 1,000 burn-in, sparsity scale $p_0 = 5$; sensitivity $p_0 = 2$ and 10 reported in Results) and report posterior means, standard deviations and the area under the ROC curve (AUC). The 5-year binary endpoint is defined only for patients evaluable at 60 months (follow-up ≥ 60 months, or relapse before 60 months); the 144 patients censored before 60 months without relapse have unknown 5-year outcome and are excluded, so the fixed-time model is a complete-case analysis with an informative-censoring caveat and is interpreted as exploratory only. Finally, to separate the contribution of the network-based gene selection from the prognostic content of any 60-gene set, we compared CS with (i) an equal-weight average of the same 60 kernel genes and (ii) two null distributions of 2,000 random 60-gene sets each (fixed seed), both drawn from the same 99-gene panel from which the kernel was selected: random genes with equal weights, and random genes with random rank-based weights. The two nulls separate the effect of the gene-selection scheme (network topology versus random choice within the same candidate panel) from the effect of the weighting scheme (centrality ranks versus random ranks). A whole-transcriptome null used in an earlier draft was dropped because it draws from a different candidate universe and is not an apples-to-apples comparison. To characterise the time structure of the association, we checked the proportional-hazards assumption with Schoenfeld residuals (R *survival::cox.zph*) on the M1 model, fitted a $CS \times \log(1 + \text{time})$ interaction model, and estimated period-specific CS hazard ratios in time splits at 24 and 60 months (*survSplit*; the few events recorded at 0 months were shifted to 0.01 month so that the left boundary lies strictly below all observed times). Time-dependent discrimination was summarised by Uno’s inverse-probability-of-censoring AUC at 1, 3, 5 and 10 years, with the censoring distribution estimated by Kaplan–Meier. Finally, we quantified the overlap of CS with established clinical covariates by Pearson correlation with ER, PR, HER2, grade and age.

2.8 Cross-Cohort Transportability Assessment in TCGA-BRCA

To assess whether the prognostic association of CS transports across cohorts, we analysed The Cancer Genome Atlas breast carcinoma cohort (TCGA-BRCA; The Cancer Genome Atlas Network 2012). Level-3 RNA sequencing expression (HiSeqV2, RSEM-normalised) and the clinical matrix were downloaded from the UCSC Xena platform (Goldman et al. 2020). Tumour samples (sample type code 1) were retained. Of the 99 genes in the METABRIC panel, 98 were present in the TCGA expression matrix (CXCL8 absent; the 60-gene kernel retained 59 of 60 genes). Per-gene expression was z-scored within TCGA, and CS was computed with the *fixed* METABRIC kernel weights c_k (Eq. 3), so the score construction used no TCGA survival information. TCGA-BRCA has no curated relapse-free-survival endpoint; we therefore assessed transportability against overall survival (OS) in days, constructed from vital status and follow-up days. The validation models replicated the METABRIC specifications: a clinical baseline (ER, PR, HER2 by immunohistochemistry, age; there is no grade variable in the TCGA clinical matrix) and a clinical + CS model. All survival statistics were computed with the R *survival* package (Cox partial likelihood, Efron tie handling; concordance; log-rank test on the median split; 5-year Kaplan–Meier estimates). This is therefore a cross-cohort transportability assessment on a different endpoint (OS rather than RFS) and a different platform (RNA-seq rather than microarray); we report it as such and discuss the endpoint and platform differences explicitly.

2.9 Cross-Cohort Network Structure Replication

Because the CS weights are derived from the METABRIC network structure, the portability of the score depends in part on whether the inferred network structure itself is reproducible in an independent cohort. We re-estimated the graphical-lasso network on the TCGA tumour samples (same 98-gene panel, same regularisation protocol: 3-fold cross-validation, six candidate penalties, $|\pi_{ij}| > 10^{-12}$ for the weighted graph and $|\pi_{ij}| > 0.05$ for the sparse graph) and compared the two edge sets by the Jaccard index on the shared gene set. This analysis is reported as a structural replication check, not as evidence about causal regulation.

All analyses were performed with Python 3.14 (numpy, scipy, scikit-learn, networkx, matplotlib) and R 4.5.2 (survival 3.8.3). The complete pipeline—data download, network inference, control-theoretic metrics, simulations, survival models and cross-cohort assessment—is available in the accompanying code repository and reproduces every figure and table in this paper.

3 Results

3.1 Cohort and Clinical Profile (METABRIC)

After quality control, 1,979 METABRIC tumours with complete RFS and expression data formed the analysis set. During follow-up, 803 relapse events were recorded (event rate 40.6%; median follow-up exceeding 10 years for censored patients). Consistent with established clinical knowledge, ER status, HER2 status and grade stratified outcome in the expected directions: the 5-year relapse rate was 22.6% among ER-positive tumours versus 40.6% among ER-negative tumours, and 43.9% among HER2-positive versus 24.6% among HER2-negative tumours; the corresponding Kaplan–Meier estimates of the 5-year relapse probability ($1 - \hat{S}(60)$) were 21.6%, 39.2%, 42.1% and 23.5%, respectively; ER-positive patients had a markedly longer median observed RFS (105.9 months) than ER-negative patients (79.0 months). These 5-year relapse rates use a KM-style denominator (patients evaluable at 60 months) and are distinct from the follow-up-wide RFS event rates reported in Table 1 (39.5% versus 22.6% for ER-positive, 43.9% versus 40.6% for ER-negative, 52.2% versus 43.9% for HER2-positive, and 38.9% versus 24.6% for HER2-negative). Table 1 summarises the clinical profile of the analysis set, and Fig. 1 shows the cohort composition by RFS event, ER status and PR status.

3.2 The Inferred GRN and its Cybernetic Organisation

Graphical-lasso inference with cross-validated regularisation ($\alpha = 0.345$) yielded a sparse conditional dependence network among the 99 regulatory genes: 317 undirected edges out of 4,851

possible pairs (density 0.065), organised into 24 connected components and 27 greedy-modularity communities, with a mean degree of 6.40 and a mean clustering coefficient of 0.37. The largest connected component contains 76 genes and all 317 edges; the remaining 23 genes (including TP53, MDM2, CDKN1A, BAX, BAD, CASP9, KRAS, BRAF, VEGFA and VEGFB) are isolated in the weighted graph. Edge stability over 25 bootstrap resamples depends on the edge-reappearance criterion. If an edge is considered present whenever its partial-correlation magnitude exceeds 10^{-12} (the threshold of the weighted graph), mean stability is high (0.947 across the 317 inferred edges) and 283 of 317 edges (89%) reappeared in more than 80% of resamples; the most stable edges were centred on the oestrogen receptor (ESR1–PGR, ESR1–AR, ESR1–TFF1, ESR1–ERBB3, ESR1–EGFR and ESR1–IGF1R, each with stability 1.00), and the ESR1–PGR edge is a textbook ER-signalling module. If edges are instead required to clear the sparsity threshold $|\pi_{ij}| > 0.05$ used for the sparse graph, stability is much lower (mean 0.442, median 0.16; only 120 of 317 edges reappeared in more than 80% of resamples), and all 25 bootstrap fits emitted scikit-learn “ConvergenceWarning” notices from the graphical-lasso estimator; we therefore treat only the aggregate topological statistics as stable and caution against over-reading any individual edge. Fig. 2 shows this largest component with the control-kernel genes highlighted.

As a cybernetic system, the inferred network exhibits substantial cyclic structure: its cyclomatic number $M = E - N + C = 317 - 99 + 24 = 242$ implies 242 independent undirected cycles, i.e. a cycle-space dimension of 2.44 per gene. In an undirected conditional-dependence graph these cycles do not identify directed regulatory feedback loops; they indicate the structural capacity of the network to support feedback wiring and closed-loop homeostasis (Wiener 1948). The linearised dynamics $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ on the weighted network have spectral radius $\rho(\mathbf{A}) = 0.904$ (largest eigenvalue 0.904, smallest -0.422 ; Fig. 3, right); with dissipation $\eta = 1.2 > \rho(\mathbf{A})$, all eigenvalues of the stabilised matrix $\mathbf{A} - \eta\mathbf{I}$ lie in the left half-plane ($\text{Re } \lambda \leq -0.296$), so the unforced linearised system is asymptotically stable: perturbations to the regulatory state decay, the dynamical signature of a homeostatic network operating near a stable operating point. For the controllability analysis, the weighted adjacency matrix has rank 75, so by the exact-controllability result for symmetric

matrices (Yuan et al. 2013) the minimum number of independent input channels is $N - \text{rank}(\mathbf{A}) = 24$, corresponding to the 24-fold zero eigenspace; this is far smaller than the naive count $N - \mu(\lambda_{\max}) = 98$ that would apply if the largest-eigenvalue multiplicity were the binding constraint, and it confirms that only the zero eigenspace (the 23 isolated genes and one extra redundant mode) is unreachable in principle. We used the top 60 genes by structural centrality (the control kernel) rather than the minimum input set to define the patient-level score, because the centrality ranking is continuous and reflects the relative importance of genes in steering the network; with these 60 inputs the reachable subspace has dimension 75 of 99, i.e. the 23 isolated, undriven genes are not controllable from the kernel inputs.

3.3 Control Energy of the Stabilised Network

For the dissipative dynamics $\dot{\mathbf{x}} = (\mathbf{A} - 1.2\mathbf{I})\mathbf{x} + \mathbf{B}\mathbf{u}$, the controllability Gramian over the 60-gene control kernel has rank 75, i.e. the kernel inputs render a 75-dimensional subspace of the 99-dimensional state space controllable. Within the reachable subspace the smallest nonzero Gramian eigenvalue is 3.06×10^{-8} and the dynamic range (largest-to-smallest nonzero eigenvalue ratio) is 5.53×10^7 ; the condition number of 1.69×10^{12} reported in the Methods uses a display floor of 10^{-12} and is therefore a conservative upper bound. The Gramian has an effective rank of 74 out of 99 modes and a maximum eigenvalue of 1.69. The 10 largest modes (10% of the spectrum) account for 25.8% of the total Gramian trace—a moderately concentrated spectrum in which a minority of modes provides the inexpensive control channels. Per-driver Gramian traces rank the control kernel in biologically sensible order: ESR1, CCNB2, BUB1, CCNA2 and CDC20 lead the list, i.e. the most controllable single-gene inputs include the oestrogen-signalling hub and core cell-cycle regulators, consistent with their established roles as breast cancer drivers. This concordance between a purely structural cybernetic ranking and prior biological knowledge is a first indication that the inferred network captures biologically meaningful regulatory organisation.

3.4 Dynamical Response and Convergence of the Network

Simulations from 40 random initial states on the inferred network were performed in two regimes. In the linear dissipative regime ($\eta = 1.2$), the horizon projections onto the Perron eigenvector of $\mathbf{A}$ span $[-5.7 \times 10^{-3}, +5.8 \times 10^{-3}]$ and split at the sample median (-5.5×10^{-4}) into a low- and a high-response half (Fig. 5, left; means -2.5×10^{-3} and $+2.3 \times 10^{-3}$, 20 initial states each). Because the linearised system has a single equilibrium, these are transient projections that decay toward the origin within the simulated horizon ($T = 12$). In the nonlinear saturating regime $\dot{\mathbf{x}} = -\mathbf{x} + \tanh(\mathbf{A}\mathbf{x}) + \mathbf{B}\mathbf{u}$, terminal projections at $t = 20$ span $[-0.027, +0.031]$ and partition into three bands by tertiles; however, multi-start fixed-point analysis of the same system located exactly one stable fixed point within the searched initial-condition space (Perron projection 0.0; largest real part of the Jacobian eigenvalues -0.096), and at $t = 50$ all 40 trajectories converge to it (projections within $[-1.6 \times 10^{-3}, +1.7 \times 10^{-3}]$; Fig. 5, right). The apparent banding is therefore a transient artefact of the finite simulation horizon rather than evidence of multistability: on this network both the linearised and the saturating dynamics are monostable, and no spontaneous phenotype switching is supported by the inferred topology. As a demonstration of input-modulated transient response, we applied a constant input $\mathbf{u}_{\text{const}} = \mathbf{B}^\top \mathbf{W}_c^+ \Delta$ (regularised Gramian; input norm 0.042) to the lowest-projection state. Within a horizon of $T = 8$, the controlled trajectory raises the Perron projection from -5.7×10^{-3} to $+3.4 \times 10^{-3}$ (integrated input energy 1.4×10^{-2} ; regularised unit-displacement energy 0.91), but the endpoint reaches 79% of the target displacement (from -5.7×10^{-3} to $+3.4 \times 10^{-3}$ against a target of $+5.8 \times 10^{-3}$) with a full-state relative error of 136% (Fig. 6): a single constant input does not complete the displacement. Moreover, the uncontrolled trajectory from the same initial state already crosses the median line at the horizon end ($T = 8$; endpoint -5.4×10^{-4}), because the stable dynamics relax toward the origin near the median projection. We therefore do not interpret this experiment as phenotype steering; it shows instead that a constant input modulates the transient trajectory, while a genuine state transfer would require time-varying optimal control in the 75-dimensional controllable subspace.

3.5 Control-Kernel Score and Relapse-Free Survival

The patient-level control-kernel score CS, defined by Eq. (3) with structure-based, outcome-blind weights on the 60 control-kernel genes, was weakly but positively correlated with the RFS event indicator ($r = 0.144$). In the Cox model M1 (CS + clinical covariates), CS was significantly associated with relapse: HR = 1.22 per SD (95% CI 1.12–1.32; $p = 5.6 \times 10^{-6}$, Table 2, Fig. 12), after adjustment for ER, PR, HER2, grade and age. A permutation test in which the CS vector was permuted 1,000 times (with survival times and event indicators held fixed) and the univariate Cox model refitted gave a permutation $p = 0.001$ (observed univariable coefficient 0.2575), confirming that the survival association is not an artefact of the score construction; this univariable statistic is distinct from the multivariable-adjusted coefficient reported in Table 2 (0.196). As a secondary exploratory analysis, a Bayesian horseshoe logistic regression of 5-year relapse on the 40 top kernel genes plus clinical covariates, restricted to the 1,835 patients evaluable at 60 months (a fixed-time, complete-case analysis: the 144 patients censored before 60 months have unknown 5-year outcome and are excluded, which introduces an informative-censoring caveat), reproduced the pattern at the level of individual genes under strong global shrinkage: the largest posterior effects were positive for AURKA (posterior mean 0.138 ± 0.130 ; posterior probability of a positive effect 0.87) and HER2+ (0.095 ± 0.068 ; 0.94), negative for BCL2 (-0.095 ± 0.092 ; 0.88 probability of a negative effect) and ESR1 (-0.079 ± 0.106 ; 0.22 probability of a positive effect), and positive but smaller for CCNB2 (0.080 ± 0.121), MKI67 (0.080 ± 0.085), grade (0.079 ± 0.077) and CDC20 (0.065 ± 0.105) (Fig. 7). All posterior means lay below 0.14 in absolute value, consistent with the strong global shrinkage implied by the horseshoe sparsity scale ($p_0 = 5$, i.e. roughly five of the 45 coefficients expected to be non-negligible; only AURKA had $P(|\beta| > 0.1) > 0.5$). The in-sample AUC of this exploratory model was 0.692, a descriptive fit summary rather than an estimate of generalisation; effective sample sizes of the post-burnin chains were moderate (minimum ESS 374, lag-1 approximation, single chain so no $\hat{R}$). The conclusions were broadly insensitive to the sparsity scale: for $p_0 = 2, 5$ and 10 the in-sample AUC was 0.692 in each case, and the top-10 features were largely stable with only marginal reordering at the tail (AURKA, BCL2, HER2+,

CCNB2, MKI67, grade(z), ESR1, MAPK1, CDC20 and ER+ for $p_0 = 5$).

One caveat on the clinical covariates deserves explicit mention: univariably, ER-positive status was associated with lower relapse hazard (HR = 0.77, 95% CI 0.66–0.90; $p = 0.001$), consistent with established clinical knowledge; after multivariable adjustment the coefficient reversed sign but remained non-significant (HR = 1.19, $p = 0.10$, Table 2). This sign reversal reflects the collinearity between receptor status and the remaining covariates in METABRIC ($\text{corr}(\text{ER}, \text{grade}) = -0.37$; $\text{corr}(\text{ER}, \text{CS}) = -0.43$). The score itself overlaps partly with established molecular phenotypes: $\text{corr}(\text{CS}, \text{grade}) = 0.51$, $\text{corr}(\text{CS}, \text{ER}) = -0.43$, $\text{corr}(\text{CS}, \text{PR}) = -0.33$, $\text{corr}(\text{CS}, \text{HER2}) = 0.21$ and $\text{corr}(\text{CS}, \text{age}) = -0.18$. CS therefore carries information that overlaps with proliferation and receptor biology, and the multivariable analysis asks whether it retains an independent association once this overlap is accounted for; we interpret the adjusted ER coefficient accordingly; the univariate estimate is the clinically informative one. A complete-case sensitivity analysis excluding the two patients with missing PR status and the one with missing HER2 status left the CS coefficient essentially unchanged (HR = 1.22, $p = 4.2 \times 10^{-6}$).

3.6 Discrimination and Robustness

Four nested levels of evidence should be distinguished. (i) *Full-cohort association*: CS was associated with RFS after adjustment (Section 3.5; HR = 1.22 per SD, $p = 5.6 \times 10^{-6}$). (ii) *Cross-validation conditional on the fixed network*: with the network, the kernel and the CS weights estimated once from the full cohort and only the Cox coefficients refitted inside each of 5×5 folds, adding CS modestly improved the Harrell C-index from 0.588 (M0) to 0.611 (M1, +0.023), with the penalised 40-gene model M2 at 0.604 (mean $\pm$ SD: M0 0.588 ± 0.015 ; M1 0.611 ± 0.017 ; M2 0.604 ± 0.018 ; Fig. 8). The within-split improvement was consistent across all 25 splits (M1 – M0 = $+0.0225 \pm 0.0040$; paired t -test $p = 2.3 \times 10^{-4}$). Because test-fold expression data participate in network inference, kernel selection and weight estimation, this protocol overstates generalisation. (iii) *Full-pipeline cross-validation*: refitting every feature-construction step—network inference, kernel selection and score weights—inside each training fold gave M0 0.587 ± 0.041 , M1

0.606 ± 0.031 and M2 0.606 ± 0.019 . These are the primary generalisation results: adding the single continuous score CS improved the C-index by 0.019 over clinical covariates alone, reproducibly across folds despite the additional uncertainty of refitting the network. The compression argument is preserved: M1 matches the 45-parameter regularised model M2 with one parameter rather than 45 (in-sample C-index 0.644 for M2 versus 0.615 for M1, confirming that the full kernel-expression model overfits more than the compressed score). (iv) *Cross-cohort transportability*: the association did not transport to TCGA-BRCA overall survival (Section 3.7). The survival difference between CS strata was strongly significant: the log-rank test (Mantel–Cox form) on the median split gave $\chi^2 = 45.5$ with $p = 1.5 \times 10^{-11}$ (Fig. 9); the 5-year Kaplan–Meier relapse-free survival was 82.6% in the CS-low stratum versus 65.8% in the CS-high stratum.

Proportional-hazards check. The proportional-hazards assumption was rejected for the M1 model (Schoenfeld global $p = 1.1 \times 10^{-21}$; CS $p = 2.0 \times 10^{-14}$), and the CS effect is strongly time-dependent. In the M1 model augmented with a $\text{CS} \times \log(1 + \text{time})$ interaction, the interaction coefficient was -0.66 (SE 0.04; $p = 4.3 \times 10^{-71}$), and period-specific HRs per SD fell from 1.65 (0–24 months; $p = 7.5 \times 10^{-13}$) to 1.29 (24–60 months; $p = 1.4 \times 10^{-4}$) and 0.91 after 60 months ($p = 0.14$). The overall HR of 1.22 is therefore a time-averaged effect; the prognostic signal is concentrated in the first five years and is no longer detectable beyond 60 months. The effect of grade remained significant in the time-split model ($p = 0.003$) while ER and age no longer were, consistent with receptor biology being partly absorbed by the time-varying structure.

Time-dependent discrimination. Uno inverse-probability-of-censoring time-dependent AUC (1, 3, 5 and 10 years) improved when CS was added to the clinical model at every horizon: 0.640 to 0.647 (1 year), 0.652 to 0.676 (3 years), 0.635 to 0.655 (5 years) and 0.594 to 0.623 (10 years), with the largest absolute gains at 3–10 years.

Kernel-size robustness. The kernel size was pre-specified as the top $K = 60$ genes ranked by the structure-based control score before any model fitting. Re-running the univariate analysis with $K = 20$ and $K = 40$ gave HRs of 1.295 and 1.291 per SD (both $p < 10^{-9}$), indistinguishable from $K = 60$ (HR 1.294; log-rank $p = 1.5 \times 10^{-11}$), so the association is not an artefact of the specific

cutoff.

Two control analyses qualify the interpretation of the network-based gene selection. An equal-weight average of the same 60 kernel genes performed broadly like the centrality-weighted CS—slightly lower in discrimination (C-index 0.602 vs 0.607; HR 1.284 vs 1.294 per SD) but with a smaller log-rank p (1.5×10^{-13} vs 1.5×10^{-11})—showing that the specific centrality-based weighting contributes little beyond a simple mean of the same genes. To separate the gene-selection effect from the weighting effect, we compared CS with two null distributions of 2,000 random 60-gene sets each (fixed seed), both drawn from the same 99-gene panel: random genes with equal weights, and random genes with random rank weights. The fractions of random sets whose log-rank p was smaller than the CS value (1.5×10^{-11}) were 13.3% (265/2,000) for the equal-weight null and 10.6% (212/2,000) for the random-rank-weight null. Both fractions are close to the CS result and neither is small: the network-based selection therefore did not show a decisive prognostic advantage over randomly sampled gene panels from the same candidate space in this cohort. The prognostic content of the kernel genes may thus reflect the well-known expression–outcome associations of many individual breast-cancer genes rather than a unique property of the inferred network, and we temper the claim that network structure per se adds prognostic information beyond any 60-gene expression panel. Finally, the Bayesian AUC of 0.692 is an in-sample quantity and should be interpreted as a descriptive robustness check rather than an estimate of generalisation performance.

3.7 Cross-Cohort Transportability Assessment in TCGA-BRCA

The assessment cohort comprised 710 TCGA-BRCA tumours with complete ER, PR, HER2, age, expression and OS information. Of the 1,101 primary tumour samples, 1,097 had expression data for 98 of the 99 panel genes and 59 of the 60 kernel genes (CXCL8 is absent from TCGA); applying a strict three-state definition of receptor status (only explicit Positive/Negative calls retained; ER missing or indeterminate in 52 samples, PR in 55, HER2 in 371), together with complete OS and age, left 710 tumours for analysis. During a median follow-up of 2.0 years (OS range 1–4,894

days), 78 deaths were recorded (event rate 11.0%). The clinical baseline model M0 (ER, PR, HER2 by immunohistochemistry, age) had a concordance index of 0.735, driven almost entirely by age (HR = 1.79 per SD, $p = 9.7 \times 10^{-8}$); ER, PR and HER2 were not significant in this cohort ($p = 0.15, 0.84$ and 0.05 , respectively). Adding CS to this model left the concordance index essentially unchanged (0.736) and the CS coefficient was not significant (HR = 1.02 per SD, 95% CI 0.80–1.29; $p = 0.88$). Univariably, CS showed no association with OS: HR = 1.00 (95% CI 0.80–1.25; $p = 0.99$) with a concordance index of 0.47 (Fig. 10; Table 3). The log-rank test on the median split was null ($\chi^2 = 0.53$, $p = 0.47$); 5-year Kaplan–Meier OS was 88.1% in the CS-low stratum versus 78.5% in the CS-high stratum, a difference consistent with chance. Subgroup analyses were similarly null: within ER-positive tumours ($n = 546$) CS had HR = 0.93 ($p = 0.58$), within ER-negative tumours ($n = 164$) HR = 0.90 ($p = 0.66$), and a tertile split of CS gave a log-rank $p = 0.58$. The prognostic association of CS observed in METABRIC therefore does not transport to TCGA-BRCA overall survival.

3.8 Cross-Cohort Network Structure Replication

Re-estimating the graphical-lasso network on the 1,097 TCGA tumour samples with expression data (cross-validated penalty $\alpha = 0.360$) yielded 355 edges in the weighted graph (versus 307 for METABRIC on the shared 98-gene panel) and 169 edges at the $|\pi_{ij}| > 0.05$ threshold (versus 135). The two edge sets overlap minimally: 28 shared edges in the weighted graph (Jaccard index 0.044) and 6 shared edges in the sparse graph (Jaccard index 0.020; Fig. 11). The inferred conditional-dependence structure is therefore strongly cohort-specific, which is expected for a regularised estimator of a high-dimensional correlation structure on different platforms, and it provides a structural explanation for the failure of the METABRIC-fitted score to transfer: the network topology—and hence the kernel weights derived from it—is not reproducible across cohorts under the present inference procedure. This negative result reinforces the epistemic distinction drawn in the Introduction: the inferred graph is an observational descriptor of one cohort and is not portable under this inference procedure; it does not by itself establish that the underlying biological regulatory

programme differs between cohorts.

4 Discussion

The central finding of this study has two parts. First, in the METABRIC cohort, the control-theoretic organisation of the breast cancer GRN—its cycle-space dimension, spectral stability and low control energy—is not merely a descriptive curiosity: a patient-level score derived purely from the network’s structural control kernel, with no access to clinical outcome, is significantly and independently associated with relapse-free survival. Second, and equally important, this association does not transport to an independent cohort on a different endpoint: in TCGA-BRCA, CS shows no association with overall survival, and the network structure from which the score is derived is not reproducible across cohorts (Jaccard 0.02–0.04). The two results together constitute the main scientific message: a structural control-theoretic score can capture prognostic signal in the cohort in which the network is inferred, but the observational network and its descriptors should not be treated as a causal, portable regulatory programme.

The cybernetic reading of the METABRIC network organises the statistical results mechanistically. The cycle-space dimension of 242 independent undirected cycles places the inferred network in a regime of dense cyclic structure whose dynamics are stable in both the linearised sense ($\rho = 0.904$ with dissipation 1.2) and the nonlinear sense (a single stable fixed point identified across the multi-start search of the saturating dynamics). Monostability, far from contradicting the cybernetic picture, is what a homeostatic regulatory system should look like: perturbations decay and the operating point is maintained, so that phenotypic switching—if it occurs—requires explicit control input rather than spontaneous fluctuation (Müller and Schuppert 2011; Zañudo et al. 2017). Developmental- landscape accounts describe phenotype transitions as moves between attractors (Feinberg and Levchenko 2023); on the network inferred here, however, multi-start fixed-point analysis located a single stable fixed point within the searched initial-condition space in both the linearised and the saturating dynamics; we therefore found no evidence of spontaneous multi-

stability in the METABRIC regulatory programme. In control-theoretic terms this is a homeostatic regime in which a single constant input can modulate the transient trajectory but cannot by itself complete a state transfer (the perturbation experiment of Section 3.4 reached 79% of the target displacement), so any substantial reorganisation of the regulatory state would require time-varying optimal control. The statistical association of CS with RFS then takes on a natural interpretation: tumours with higher expression across the genes of the control kernel had higher relapse risk, and this association is captured by the weighted expression score. We emphasise that CS is a weighted linear mean of 60 gene expressions; it does not directly measure displacement of a regulatory state from an attractor.

The concordance between structure-based rankings and biology strengthens this interpretation within METABRIC. The most stable network edges centre on ESR1; the most controllable single-gene inputs are ESR1 and the cell-cycle regulators (CCNB2, BUB1, CCNA2, CDC20); and the Bayesian posterior identifies AURKA, BCL2, HER2+, grade and CCNB2 among the features most strongly associated with 5-year relapse. The structure-based rankings (edge stability, per-driver control energy) and the outcome-based regularised regression thus overlap within the top ten features on CCNB2, ESR1 and CDC20, with CCNB2 the most robust—genes with established roles in oestrogen signalling and the cell cycle in breast cancer. We view this overlap as evidence that the graphical-lasso network recovers meaningful gene-association structure in this cohort rather than mere correlational noise, while noting that the overlap is partial and the outcome-based effect sizes are modest.

Why does the association fail to transport to TCGA-BRCA? Several non-exclusive explanations are supported by the data. (i) *Endpoint difference*: METABRIC RFS records relapse events (803 events), whereas TCGA-BRCA here supports only an OS endpoint (78 events). Relapse and overall survival are differently composed outcomes: OS is diluted by deaths from other causes and by treatment effects after relapse, and a score tuned to relapse biology need not track OS. (ii) *Platform and cohort differences*: METABRIC is microarray-based, TCGA is RNA-seq based, and the two cohorts differ in treatment era and case mix; the 98-gene panel overlaps but the inferred

conditional-dependence structures do not (Jaccard 0.02–0.04), so the kernel weights estimated on METABRIC are not anchored in reproducible TCGA structure. (iii) *Statistical power*: with 78 events versus 803, the TCGA analysis has substantially lower power; the point estimate (HR $\approx$ 1.00) shows no trend in the hypothesised direction, and the 95% confidence interval (0.80–1.29) leaves little room for an effect as large as the METABRIC estimate (1.22) while a modest association cannot be excluded. We therefore conclude that there is no evidence of a METABRIC-sized association, not that no association of any size exists. (iv) *Score specification*: CS is a linear, centrally weighted expression summary; the equal-weight control analysis in METABRIC showed that the specific weighting scheme is not the source of the signal, and a different score construction—or a TCGA-refitted network—might behave differently. We do not claim to distinguish these explanations; we report the null result in full because it delineates the boundary of the method.

Several limitations should be stated explicitly. First, the network is undirected and inferred from cross-sectional expression; it captures conditional dependence, not causation, and the controllability analysis is structural in the sense of the exact-controllability formalism (Yuan et al. 2013) rather than a claim about experimentally achievable perturbations. Second, the 60 kernel inputs reach only a 75-dimensional controllable subspace: the 23 isolated genes (including TP53, MDM2, CDKN1A and the BCL2-family pro-apoptotic genes BAX and BAD) are not directly driven by the kernel, so the score characterises the reachable regulatory core rather than the full network; the minimum-input count of 24 refers to the zero eigenspace and does not imply that breast cancer regulatory programmes are experimentally easy to steer—the control-energy spectrum is the more informative descriptor (Yan et al. 2015). Third, the prognostic analysis rests on a single training cohort; the TCGA-BRCA cross-cohort assessment shows that the association is cohort- and endpoint-specific, and the score is not currently a candidate biomarker. The in-sample Bayesian AUC, the large fold-to-fold SD of the full-pipeline cross-validation, and the control analyses showing that the equal-weight score performs as well as the centrality-weighted score, and that 13.3% and 10.6% of random 60-gene sets (two null distributions within the same 99-gene panel) outperform CS in the log-rank comparison, all caution against over-interpreting the

network-based gene selection and the specific weighting scheme. Fourth, expression was z-scored within each cohort and the score definition depends on the inferred network; both steps are fully deterministic and reproducible, but the score’s transportability across platforms is not supported by the present data. Finally, the METABRIC RFS endpoint is registry-defined and includes both local and distant relapses; analyses of distant-relapse-free survival, overall survival or subtype-specific strata may reveal different structure–effect relationships—indeed, the TCGA-BRCA OS result is a direct demonstration that they do.

Notwithstanding these limitations, the present study demonstrates a methodological point that we believe is of general value: descriptors of network structure that originate in control theory (cycle-space dimension, spectral radius, controllability, control energy) can be translated into patient-level biostatistical covariates in a fully outcome-blind manner, and at least one such descriptor carries incremental prognostic information for relapse in one large cohort. The same construction can in principle be applied to other cohorts and tumour types; whether it retains its prognostic value there is an empirical question, and the present cross-cohort assessment shows that the answer is not guaranteed.

5 Conclusions

The breast cancer gene network inferred from 1,979 METABRIC tumours admits a formal control-theoretic description: it is cycle-rich (242 undirected cycles), stable under a dissipative realisation ($\rho = 0.904$, dissipation 1.2), dynamically monostable (a single stable fixed point identified across the multi-start search) and reachable only in part—the 60 control-kernel inputs span a 75-dimensional controllable subspace (minimum independent input channels 24), and a constant-input perturbation modulates transient response without completing a state transfer (input norm 0.042; regularised unit-displacement energy 0.91; endpoint projection $+3.4 \times 10^{-3}$ versus target $+5.8 \times 10^{-3}$). A patient-level score built from the structural control kernel of this network, with no access to clinical outcome, is significantly associated with relapse-free survival (HR = 1.22

per SD, $p = 5.6 \times 10^{-6}$) after adjustment for standard clinical covariates, with strongly separated survival strata (log-rank $\chi^2 = 45.5$, $p = 1.5 \times 10^{-11}$) and improved full-pipeline cross-validated discrimination (C-index $0.587 \pm 0.041 \rightarrow 0.606 \pm 0.031$; conditional on the fixed network, $0.588 \rightarrow 0.611$), with an effect concentrated in the first five years after diagnosis. The cross-cohort assessment in TCGA-BRCA, however, showed no association of the score with overall survival (HR = 1.00, 95% CI 0.80–1.25, $p = 0.99$) and no reproducible network structure across cohorts (Jaccard 0.02–0.04). These findings demonstrate that outcome-blind network descriptors can be translated into patient-level prognostic features within a large breast cancer cohort, while the lack of cross-cohort replication and the undirected, observational nature of the inferred network caution against interpreting it as a causal, portable regulatory programme. The analysis also illustrates a general methodological point: the complex-adaptive-system framing of a gene network does not automatically imply multistability, and the limits of transferring structural network scores across cohorts and endpoints must be established empirically rather than assumed.

Author Contributions

The author conceived the study, designed and implemented all analyses, and wrote the manuscript.

Author Disclosure Statement

No competing financial interests exist.

Funding Information

No funding was received for this study.

Ethical Considerations

This study used only de-identified, publicly available data (METABRIC, TCGA-BRCA); no ethics approval was required.

Consent to Participate

Not applicable (secondary analysis of de-identified public data).

Consent for Publication

Not applicable.

Data and Code Availability

The METABRIC expression and clinical data are publicly available from the cBioPortal repository (study `brca_metabric`); the TCGA-BRCA expression and clinical data are publicly available from the UCSC Xena platform; data retrieval and pre-processing are documented in the README of the Supplementary Material. All analysis code (18 Python and R scripts), the generated figures and the complete set of result tables are provided as Supplementary Material (JCB_Supplementary.zip). A DOI-stamped archived copy of the code will be deposited in Zenodo upon acceptance, and every figure and table can be reproduced by running the pipeline end to end.

References

Berner, R., Gross, T., Kuehn, C. et al. (2023) Adaptive dynamical networks. *Phys. Rep.*, 1031, 1–59.

Cox, D.R. (1972) Regression models and life-tables. *J. R. Stat. Soc. Ser. B*, 34, 187–220.

- Curtis, C., Shah, S.P., Chin, S.F. et al. (2012) The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups. *Nature*, 486, 346–352.
- Feinberg, A.P. and Levchenko, A. (2023) Epigenetics as a mediator of plasticity in cancer. *Science*, 379, eaaw3835.
- Friedman, J., Hastie, T. and Tibshirani, R. (2008) Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9, 432–441.
- Gao, J., Liu, Y.Y., D’Souza, R.M. et al. (2014) Target control of complex networks. *Nat. Commun.*, 5, 5415.
- Goldman, M.J., Craft, B., Hastie, M. et al. (2020) Visualizing and interpreting cancer genomics data via the Xena platform. *Nat. Biotechnol.*, 38, 675–678.
- Hatzis, C., Pusztai, L., Valero, V. et al. (2011) A genomic predictor of response and survival following taxane-anthracycline chemotherapy for invasive breast cancer. *JAMA*, 305, 1873–1881.
- Huo, Y., Li, X., Xu, P. et al. (2022) Analysis of breast cancer based on the dysregulated network. *Front. Genet.*, 13, 856075.
- Li, C. and Balázsi, G. (2018) A landscape view on the interplay between EMT and cancer metastasis. *npj Syst. Biol. Appl.*, 4, 34.
- Liu, Y.Y. and Barabási, A.L. (2016) Control principles of complex systems. *Rev. Mod. Phys.*, 88, 035006.
- Liu, Y.Y., Slotine, J.J. and Barabási, A.L. (2011) Controllability of complex networks. *Nature*, 473, 167–173.
- Müller, F.J. and Schuppert, A. (2011) Few inputs can reprogram biological networks. *Nature*, 478, E4–E5.
- Nersisyan, S., Galatenko, A., Galatenko, V. et al. (2021) miRGTF-net: integrative miRNA-gene-TF network analysis reveals key drivers of breast cancer recurrence. *PLoS One*, 16, e0249424.
- Rueda, O.M., Sammut, S.J., Seoane, J.A. et al. (2019) Dynamics of breast-cancer relapse reveal late-recurring ER-positive genomic subgroups. *Nature*, 567, 399–404.
- Sgariglia, D., Conforte, A.J., Pedreira, C.E. et al. (2021) Data-driven modeling of breast cancer

tumors using Boolean networks. *Front. Big Data*, 4, 656395.

Shin, D. and Cho, K.H. (2023) Critical transition and reversion of tumorigenesis. *Exp. Mol. Med.*, 55, 692–705.

The Cancer Genome Atlas Network (2012) Comprehensive molecular portraits of human breast tumours. *Nature*, 490, 61–70.

Thomas, P.J., Olufsen, M., Sepulchre, R. et al. (2019) Control theory in biology and medicine. *Biol. Cybern.*, 113, 1–6.

Wang, Y. and Liu, Z.P. (2022) Identifying biomarkers for breast cancer by gene regulatory network rewiring. *BMC Bioinformatics*, 22, 308.

Wiener, N. (1948) *Cybernetics: Or Control and Communication in the Animal and the Machine*. Cambridge, MA: MIT Press.

Yan, G., Tsekenis, G., Barzel, B. et al. (2015) Spectrum of controlling and observing complex networks. *Nat. Phys.*, 11, 779–786.

Yuan, Z., Zhao, C., Di, Z. et al. (2013) Exact controllability of complex networks. *Nat. Commun.*, 4, 2447.

Zañudo, J.G.T., Yang, G. and Albert, R. (2017) Structure-based control of complex networks with nonlinear dynamics. *Proc. Natl. Acad. Sci. USA*, 114, 7234–7239.

Tables

Figure Legends

Figure 1. Cohort composition of the METABRIC analysis set: proportion of RFS events (40.6%), ER status (ER+ 76.0%, ER– 24.0%) and PR status (PR+ 52.5%, PR– 47.5%, PR missing 0.1%, i.e. 2 of 1,979 patients).

Figure 2. Largest connected component of the inferred breast-cancer conditional-dependence network (full weighted network, $|\rho_{ij}| > 10^{-12}$): 76 of 99 genes and all 317 edges. Node colours

Table 1: Clinical profile of the METABRIC analysis set ($n = 1,979$). RFS event rate is the overall (follow-up-wide) relapse proportion; 5-year relapse rates are reported separately in Section 3.1.

Characteristic	n	RFS event rate
ER-positive	1,505	39.5%
ER-negative	474	43.9%
PR-positive	1,038	38.2%
PR-negative	939	43.2%
PR missing	2	–
HER2-positive	247	52.2%
HER2-negative	1,731	38.9%
HER2 missing	1	–
Grade 1	169	24.3%
Grade 2	771	38.0%
Grade 3	951	46.1%

Table 2: Cox proportional-hazards model M1 (CS + clinical covariates) for relapse-free survival in METABRIC. HRs are per SD for continuous covariates; ER, PR and HER2 are binary (positive versus negative).

Variable	$\hat{\beta}$ (SE)	HR	95% CI	p
CS (per SD)	0.196 (0.043)	1.22	1.12–1.32	5.6×10^{-6}
ER+	0.175 (0.107)	1.19	0.97–1.47	0.10
PR+	−0.078 (0.086)	0.93	0.78–1.10	0.37
HER2+	0.340 (0.101)	1.41	1.15–1.71	7.4×10^{-4}
Grade (per SD)	0.130 (0.045)	1.14	1.04–1.24	3.6×10^{-3}
Age (per SD)	−0.009 (0.038)	0.99	0.92–1.07	0.82

encode greedy-modularity communities; red nodes are the control-kernel genes shown (top 25 by structural centrality). The network is organised around the oestrogen receptor (ESR1) and cell-cycle modules. The remaining 23 genes are isolated in the weighted graph and are not directly driven by the kernel inputs.

Figure 3. Left: degree distribution of the inferred gene network. Right: spectrum of the weighted adjacency matrix $\mathbf{A}$. With dissipation $\eta = 1.2 > \rho(\mathbf{A}) = 0.904$, every eigenvalue of the stabilised matrix $\mathbf{A} - \eta\mathbf{I}$ lies in the left half-plane ($\text{Re } \lambda \leq -0.296$), so the linearised dissipative dynamics are locally stable.

Figure 4. Control-energy spectrum of the stabilised network: eigenvalues of the controllability Gramian $\mathbf{W}_c$ versus rank (log scale). Within the reachable subspace the spectrum spans about eight

Table 3: Cross-cohort assessment of the control-kernel score in TCGA-BRCA (n = 710; 78 overall-survival events, strict three-state receptor calls). CS HRs are per SD. M0: clinical baseline (ER, PR, HER2 by immunohistochemistry, age). Concordance indices are from the fitted Cox models.

Model	HR (95% CI)	<i>p</i>	C-index
CS (univariable)	1.00 (0.80–1.25)	0.99	0.47
M0 (clinical)	–	–	0.735
M1 (clinical + CS)	1.02 (0.80–1.29)	0.88	0.736
Log-rank (median split)	–	0.47	–
5-year OS: CS-low 88.1% vs. CS-high 78.5%			

orders of magnitude (smallest nonzero eigenvalue 3.06×10^{-8} ; effective rank 74); the 10 largest modes (10% of the spectrum, shaded band at ranks 90–99) carry 25.8% of the total Gramian trace.

Figure 5. Dynamical response of the inferred network to 40 random initial states. Left: linear dissipative regime; transient Perron projections at the simulation horizon split at the sample median into low- and high-response halves (median line). Right: nonlinear saturating regime; terminal projections at $t = 20$ (grey) converge to the single stable fixed point identified by multi-start search by $t = 50$ (red, Perron projection ≈ 0), showing monostable convergence rather than coexisting attractor bands.

Figure 6. Constant-input perturbation of the network via the control kernel (input norm 0.042). The controlled trajectory (solid) raises the transient Perron projection from the lowest state toward the high-response pole, while the uncontrolled trajectory (dashed) from the same initial state relaxes toward the origin and crosses the median line only at the horizon end ($T = 8$). The endpoint of the controlled run ($+3.4 \times 10^{-3}$) remains below the target projection ($+5.8 \times 10^{-3}$; full-state relative error 136%), so the experiment demonstrates input-modulated transient response rather than phenotype steering.

Figure 7. Bayesian horseshoe logistic regression of 5-year relapse on the 40 control-kernel genes plus clinical covariates, restricted to the 1,835 patients evaluable at 60 months. Shown are the top standardised coefficients by absolute posterior mean (error bars: posterior mean ± 2 SD). AURKA and HER2+ are positive, BCL2 and ESR1 negative, and CCNB2, MKI67 and grade positive; all absolute posterior means are below 0.14, reflecting the strong global shrinkage of the

horseshoe prior ($p_0 = 5$).

Figure 8. Prognostic discrimination of the three nested Cox models under repeated 5-fold cross-validation (25 splits, features fixed from the full-cohort analysis): M0 (clinical only) 0.588, M1 (clinical + CS) 0.611, M2 (Ridge-Cox on 40 kernel genes) 0.604. Error bars: SD across splits; dashed line: 0.5 reference.

Figure 9. Kaplan–Meier estimates of relapse-free survival for CS-high versus CS-low strata (median split) in METABRIC. The strata separate strongly (log-rank $\chi^2 = 45.5$, $p = 1.5 \times 10^{-11}$); 5-year relapse-free survival is 82.6% (CS-low) versus 65.8% (CS-high).

Figure 10. Kaplan–Meier estimates of overall survival for CS-high versus CS-low strata (median split) in TCGA-BRCA ($n = 710$). The strata do not separate (log-rank $\chi^2 = 0.53$, $p = 0.47$); 5-year overall survival is 88.1% (CS-low) versus 78.5% (CS-high).

Figure 11. Cross-cohort replication of the inferred network structure on the shared 98-gene panel: number of edges in the METABRIC (training) and TCGA-BRCA (external) networks and their overlap, for the dense ($|\pi_{ij}| > 10^{-12}$; left) and sparse ($|\pi_{ij}| > 0.05$; right) thresholds. Jaccard indices are 0.044 and 0.020, respectively.

Figure 12. Forest plot of the multivariable Cox model M1 (CS + clinical covariates) for relapse-free survival in METABRIC, sorted by hazard ratio. CS per SD is associated with a hazard ratio of 1.22 (95% CI 1.12–1.32); HER2-positive status is the only clinical covariate with a comparable effect size in this model.

Figures











